# Taiwan Semiconductor Manufacturing Co (TSM)

## The AI Infrastructure Inflection Point Sets the Stage for a Measured Re-rating

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> Taiwan Semiconductor Manufacturing Co continues to benefit from durable demand for leading-edge logic while the market overemphasizes near-term cyclical softness in consumer electronics. Our analysis shows that the company's fundamentals support a blended target price of $244.78, implying -42.0% upside from the current reference level.

| HOLD | $244.78 | -42.0% | $421.69 |
|---|---|---|---|
| Hold, Valuation Concerns | BLENDED TARGET | BLENDED UPSIDE | REFERENCE PRICE |
| INVESTMENT RATING | | | |

*Investors are fixated on quarter-to-quarter wafer-start fluctuations, yet the multi-year compounding power of TSMC's technology leadership and captive customer base remains the dominant driver of intrinsic value.*

## EXECUTIVE SUMMARY

Taiwan Semiconductor Manufacturing Co operates the world's most advanced semiconductor foundry, with a portfolio that spans mature nodes, specialty technologies and the latest 3-nanometer and 2-nanometer processes. The company serves a diversified client base that includes fabless designers, integrated device manufacturers and emerging AI chip developers, giving it exposure to multiple growth vectors. Its scale enables it to capture over 55% of global foundry revenue, a share that has been stable despite increased competition from Samsung and Intel. The recent share price decline reflects a short-term overreaction to softer demand in smartphone and personal computer segments, which has caused investors to discount the longer-term secular tailwinds from artificial intelligence, high-performance computing and automotive electronics. While near-term utilization rates have dipped, TSMC's capacity expansion plan remains on track, and its pricing power at the leading edge has shown resilience. Capital allocation remains disciplined, with free cash flow consistently reinvested into next-generation fab construction, advanced packaging and research initiatives. The balance sheet is robust, featuring net cash exceeding $30 billion and a debt-to-equity ratio below 0.2, providing ample flexibility to pursue selective M&A or to weather cyclical downturns without compromising its technological lead.

## OPERATIONAL SNAPSHOT

| $421.69 | $95.0B | $23.0B | 24% | 1.44 | $421.69 |
|---|---|---|---|---|---|
| REF. PRICE | LTM REVENUE | LTM FREE CASH FLOW | FCF MARGIN | EQUITY BETA | REF. CLOSE |

*LTM figures are Cascadia estimates grounded in the latest filings; ref. price as of 2026-10-15.*

## THE MOAT IS DURABLE AND EXPANSIVE

TSMC’s switching costs are exceptionally high because customers invest years in design-for-manufacturing collaboration, mask sets and process qualification that are non-transferable to alternative foundries. This creates a durable lock-in that protects pricing power even when alternative capacity becomes available. Regulatory barriers further reinforce the moat, as the construction of a leading-edge fab requires multi-year government approvals, massive capital commitments and access to scarce specialty materials, deterring rapid entry by new competitors. The company’s extensive patent portfolio, covering lithography, etch and deposition techniques, adds another layer of protection that rivals cannot easily circumvent. Customer retention rates exceed 90% for the top-tier clients, reflecting the trust built through consistent yield improvements and on-time delivery. This loyalty translates into predictable revenue streams and allows TSMC to negotiate favorable long-term supply agreements. Pricing power remains evident at the leading edge, where TSMC can command premiums of 10-15% over legacy nodes due to the performance and power efficiency advantages its technology delivers. The combination of these factors creates a moat that is both durable in its defensive characteristics and expansive in its ability to capture value across multiple end markets.

## VALUATION

### Three scenarios make the valuation range explicit

Our baseline revenue compounding assumes a mid-single-digit CAGR driven by steady adoption of AI accelerators and continued expansion of automotive semiconductor demand, while normalized free cash flow margins gravitate toward the historical mid-20% range as capacity utilization stabilizes. Terminal value is calculated using a perpetual growth rate consistent with long-term GDP expansion, and the discount rate reflects the live WACC derived from the CAPM framework.

| SCENARIO | WEIGHT | REVENUE CAGR | TARGET FCF MARGIN | TARGET | UPSIDE |
|---|---|---|---|---|---|
| Bear | 25% | 2.0% | 20% | $199.97 | -52.6% |
| Base | 50% | 4.0% | 24% | $243.06 | -42.4% |
| Bull | 25% | 6.0% | 28% | $293.04 | -30.5% |

**Bear $199.97**

In the bear scenario, a prolonged slowdown in consumer electronics keeps wafer-start growth below 2% annually, while competitive pressure forces TSMC to accept lower pricing at the leading edge, compressing free cash flow margins to 20%.

**Base $243.06**

The base case assumes a balanced environment where AI-driven demand offsets cyclical weakness, delivering a 4% revenue CAGR and allowing margins to normalize at the historical 24% level as utilization returns to long-term averages.

**Bull $293.04**

Under the bull scenario, rapid adoption of generative AI and edge computing fuels a 6% revenue CAGR, while TSMC's pricing power and improved mix lift free cash flow margins toward 28% as higher-value nodes contribute a larger share of sales.

## VALUATION FRAMEWORK

### A live discount rate, not a static hurdle

We derive the 9.84% WACC using the CAPM formula with a risk-free rate of 3.5%, an equity risk premium of 5.5% and a beta of 1.44 that reflects TSMC's exposure to global technology cycles. This rate implies that the market prices the stock as if it bears a moderate-to-high level of systematic risk, consistent with its cyclical yet structurally advantaged profile.

### FCF margin normalization is the engine

Across the scenarios, free cash flow margins move from a depressed 20% in the bear case to a robust 28% in the bull case, driven primarily by changes in utilization, product mix and pricing power at the leading edge. The base case assumes a return to the historical average margin as the industry works off excess capacity and demand stabilizes.

### Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A and stock compensation, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

**Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

**KEY VALUATION PARAMETERS**

| | |
|---|---|
| Revenue base year | **2026** |
| Forecast horizon | **10 years** |
| Live WACC | **9.84% (2026-10-15)** |
| Terminal growth | **3.0%** |
| Equity beta | **1.44** |
| Diluted shares | **25.9B** |

**HOLD**

**over $244.78 blended target**

## BUSINESS & SEGMENT DRIVERS

### Revenue prospects anchor on AI accelerators and advanced nodes HOLD

TSMC’s revenue is increasingly tied to the proliferation of AI accelerators, which require the latest 3-nanometer and forthcoming 2-nanometer nodes to achieve the performance and power efficiency demanded by data-center and edge applications. This segment has grown at a compound annual rate exceeding 20% over the past three years and now represents roughly 30% of total wafer revenue. Advanced nodes, defined as 5-nanometer and finer, continue to expand beyond AI into high-performance computing, 5G infrastructure and automotive systems, providing a diversified growth base. The company’s capacity at these nodes is expected to increase by approximately 40% over the next five years as new fabs in Arizona, Japan and Germany come online. Mature nodes, while contributing a smaller share of revenue growth, remain important for cash flow generation due to their higher utilization and lower capital intensity. TSMC’s ability to shift wafer starts between mature and advanced nodes based on demand signals provides operational flexibility that supports steady free cash flow generation. Packaging and test services, particularly advanced 2.5D and 3D solutions, are emerging as a higher-margin adjunct to front-end fabrication, driven by the need to integrate heterogeneous chiplets for AI and high-performance computing applications.

**What to monitor**

- **Capital intensity trends** Changes in the ratio of capital expenditures to revenue will signal whether TSMC is over-investing in capacity relative to demand.
- **Customer concentration risk** Monitoring the share of revenue attributable to the top five clients helps assess vulnerability to shifts in key accounts.
- **Geopolitical exposure** Tracking developments in U.S.–China trade policy and regional security situations is essential given TSMC’s global fab footprint.
- **Technology roadmap execution** Evaluating yield ramp and timing for the 2-nanometer node provides insight into the company’s ability to maintain its technological lead.

| 5.0% | 53% | 30% | 8% | 5.5% | 85% |
|---|---|---|---|---|---|
| REVENUE YOY GROWTH | GROSS MARGIN | CAPEX INTENSITY | R&D INTENSITY | FREE CASH FLOW YIELD | CAPACITY UTILIZATION |

| METRIC | LATEST FY |
|---|---|
| Revenue | $92.0B |
| Gross Profit | $48.8B |
| EBIT | $22.0B |
| Net Income | $18.0B |
| Free Cash Flow | $22.0B |

| | |
|---|---|
| R&D | $8.0B |
| ROE | 18% |

## RISK & MOAT MATRIX

### The bear case

**SHARE EROSION**

Intensifying competition from Samsung Foundry and Intel's IDM 2.0 strategy could pressure TSMC's market share, particularly in the mid-range node segment where capacity overlaps are greatest.

**M&A DIGESTION**

Any large-scale acquisition of specialty packaging or software assets would require successful integration to avoid diluting returns on invested capital.

**REGULATORY AND PRICING SCRUTINY**

Governments may impose export controls or pricing investigations on leading-edge wafers, potentially limiting TSMC's ability to capture premiums in certain jurisdictions.

**TECHNOLOGY DISRUPTION**

A breakthrough in alternative lithography, such as directed self-assembly or novel etching techniques, could reduce the barriers to entry for new foundry entrants.

**REINVESTMENT TREADMILL**

Sustaining technological leadership demands continuous multi-billion-dollar fab investments, which could strain free cash flow if demand fails to keep pace with capacity additions.

**WACC AND DURATION SENSITIVITY**

Given the long-dated nature of TSMC's cash flows, a rise in long-term interest rates would increase the discount rate and materially lower the present value of future free cash flow.

### The strengths

**OLIGOPOLY POSITION**

TSMC, Samsung and Intel collectively control over 80% of global foundry capacity, creating an oligopolistic environment that limits price-based competition and supports stable margins.

**FCF COMPOUNDING**

The company's ability to generate free cash flow in excess of its cost of capital enables reinvestment into next-generation nodes, creating a self-reinforcing cycle of technological

**GEOGRAPHIC DURABILITY**

TSMC's fab diversification across Taiwan, the United States, Japan and Europe reduces exposure to any single-point geopolitical shock while preserving

**PIPELINE OPTIONALITY**

A deep technology roadmap spanning nodes from 28-nanometer to 2-nanometer provides strategic flexibility to shift wafer mixes in response to evolving end-market

| | | | |
|---|---|---|---|
| | advancement and shareholder value creation. | access to key talent pools and supply chains. | demand without sacrificing performance leadership. |

### Compelling risk-reward, skewed to the upside

While near-term headwinds in consumer electronics have prompted a valuation compression, the structural advantages of TSMC’s durable moat, its entrenched position in the AI accelerator supply chain and its disciplined capital allocation framework suggest that the market is undervaluing the company’s long-term compounding potential. Our probability-weighted DCF analysis yields a blended target price of $244.78, which represents -42.0% upside relative to the current reference price of $421.69. Accordingly, we maintain a Hold rating with a valuation-concerns qualifier, reflecting the view that the stock is fairly priced given the prevailing risk-return trade-off.

**Justin Wen, Senior Equity Research Analyst, Cascadia Analytics**

## FRANCHISE DEEP DIVE

### N3 node: repriced on normalization

The N3 (3-nanometer) process node has been a cornerstone of TSMC’s recent growth, delivering significant performance and power-efficiency gains that are critical for AI accelerators, high-performance CPUs and next-generation graphics processors. Initial adoption was rapid, driven by early-access agreements with major fabless designers who sought the performance edge for flagship products. As the node matures, the initial premium pricing associated with early-access and low-volume production has begun to normalize, reflecting higher yields, increased competition from alternative architectures and the natural diffusion of the technology across a broader customer base. This normalization is evident in the sequential decline of average selling prices for N3 wafers observed over the past two quarters, even as wafer-start volumes continue to rise. The shift from a scarcity-driven pricing environment to a more balanced market does not indicate a loss of technological leadership; rather, it signals the successful transition of the N3 node from a niche, high-margin offering to a mainstream, volume-driven product that underpins the company’s long-term revenue base.

**Why the growth numbers disappointed**

- Wafer-start growth slowed to 3% quarter-over-quarter as consumer-device demand softened.
- Average selling prices for N3 declined approximately 8% due to increased competition and higher yields.
- Capital expenditures rose sharply to support new fab construction, pressuring free cash flow conversion.
- Geopolitical export controls limited shipments to certain Chinese customers, creating a temporary revenue headwind.

**Our read:** Normalization, not share loss. The moderation in N3 pricing and wafer-start rates reflects the natural maturation of a leading-edge node rather than a fundamental erosion of TSMC’s competitive position or a break in its growth thesis.

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